CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

LECTURE PLAN

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	**Program**	**Master of Computer Applications (MC305)**
Master Subject Coordinator Name:	Mandeep Singh	**Master Subject Coordinator E-Code:**	**E8730**
Course Name	Advanced Database Management System	**Course Code**	**25CAT-601**

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
3	0	0	0	3.00	T

Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1. To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2. To provide a framework through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3. To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4. To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5. To undertake societal activities for upliftment of rural/deprived sections of the society.
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)	
PEO1	Establish a well-fortified computing foundation of successful professionals by applying computing fundamentals and domain-specific knowledge, demonstrating their innovative skills and considering social and environmental concerns.
PEO2	Undertake successful implementation of ethical solutions as an individual or a member or a leader of a team by investigating, analyzing, formulating and solving complex computing problems in multidisciplinary approaches using modern tools.
PEO3	Enhance professionalism and ethical attitude in the profession while communicating with local, national, and foreign peers, bound within regulations and leading to lifelong learning.
PEO4	Promote awareness for uplifting health, safety, legal, environmental, ethical and cultural diversity issues for serving the society.

Program Specific OutComes(PSOs)	
PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)	
PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Fundamentals of Database Systems	Elmasri & Navathe	Fourth Edition	Pearson	2004
2	Introduction to Database Management System	C. J. Date	Eighth Edition	Addison-Wesley	2003
3	Database Management Systems	Raghu Ramakrishnan and Johannes Gehrke	Third Edition	McGraw-Hill Education	2002
4	Database System Concepts	Abraham Silberschatz, Henry F. Korth, and S. Suda	Seventh edition	McGraw-Hill Education	2019

Reference Books

Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	An Introduction to Database Management Systems	Bipin C. Desai	Revised Edition	Galgotia Publications Pvt Ltd	2013
2	Principles of distributed database.	M. Tamer Ozsu	Forth edition	Springer	2019

Course OutCome

SrNo	OutCome
CO1	Express the basic concepts of DBMS and RDBMS.
CO2	Apply normalization theory to the normalization of a database
CO3	Apply the concept of Transaction Management & Recovery techniques in RDBMS.
CO4	Analyze various advanced databases prevailing in market, Big Data, Temporal Databases
CO5	Parallel and Distributed Databases, XML Database and multidimensional Databases

Lecture Plan Preview-Theory

Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	Introduction to Data Base Management System	Architecture of DBMS	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	2	Introduction to Data Base Management System	Components of DBMS	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	3	Introduction to Data Base Management System	Data base models	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1

1	4	Introduction to Data Base Management System	Normalization ,Define DBLC	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
1	5	Introduction to Data Base Management System	Normalization ,Define DBLC	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO2
1	6	Introduction to Data Base Management System	Phases of DBLC.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	7	Distributed Database Management System	Introduction to DDBMS, Types of DDBMS	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1

1	8	Distributed Database Management System	Different Levels of Data & Process Distribution-SPSD	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	9	Distributed Database Management System	Different Levels of Data & Process Distribution-MPSD MPMD	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	10	Distributed Database Design	Top-down approach	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	11	Distributed Database Design	fragmentation of DB	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1

1	12	Distributed Database Design	fragmentation of DB	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	13	Distributed Database Design	Designing issues	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	14	Distributed Database Design	Designing issues	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
1	15	Distributed Database Design	Data Allocation	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1

2	16	Database Integration & Access Control	Bottom-up Design Approach, schema matching, schema integration, schema mapping	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO1
2	17	Database Integration & Access Control	data cleaning, View management, data security, semantic Integrity control.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
2	18	Database Integration & Access Control	data cleaning, View management, data security, semantic Integrity control.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
2	19	Database Integration & Access Control	View management, data security, semantic Integrity control.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3

2	20	Query processing	Query processing problem, objectives of query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	21	Query processing	Query processing problem, objectives of query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	22	Query processing	relational algebra operations, characterization of query processors	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	23	Query processing	relational algebra operations, characterization of query processors	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

2	24	Query processing	layers of query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	25	Query processing	layers of query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	26	Query decomposition and data localization	Query decomposition, localization of distributed data, query optimization	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	27	Query decomposition and data localization	Query decomposition, localization of distributed data, query optimization,	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

2	28	Query decomposition and data localization	join ordering in distributed queries, distributed query optimization. Issues in multi database query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	29	Query decomposition and data localization	join ordering in distributed queries, distributed query optimization. Issues in multi database query processing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
2	30	Query decomposition and data localization	query rewriting using views, query optimization and execution.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	31	Transaction management and concurrency control	Introduction to transaction, properties, types, architecture. Serializability, locking based concurrency control algorithms	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3

3	32	Transaction management and concurrency control	Introduction to transaction, properties, types, architecture. Serializability, locking based concurrency control algorithms	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	33	Transaction management and concurrency control	Introduction to transaction, properties, types, architecture. Serializability, locking based concurrency control algorithms	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	34	Transaction management and concurrency control	timestamp-based concurrency control, optimistic concurrency control, deadlock management.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	35	Transaction management and concurrency control	timestamp-based concurrency control, optimistic concurrency control, deadlock management.	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3

3	36	Recovery Techniques	Failure Classification in DBMS, Storage, Recovery and Atomicity, Recovery Algorithm	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	37	Recovery Techniques	Failure Classification in DBMS, Storage, Recovery and Atomicity, Recovery Algorithm	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	38	Recovery Techniques	Failure Classification in DBMS, Storage, Recovery and Atomicity, Recovery Algorithm	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO3
3	39	Enhanced Data Models for Advanced Applications	Active Database Concepts and Triggers	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

3	40	Enhanced Data Models for Advanced Applications	Active Database Concepts and Triggers	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	41	Enhanced Data Models for Advanced Applications	Temporal Database Concepts, Spatial Database Concepts	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	42	Enhanced Data Models for Advanced Applications	Spatial Database Concept	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO4
3	43	Enhanced Data Models for Advanced Applications	Temporal Database Concepts, Spatial Database Concepts	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

3	44	Streaming data and cloud computing	Management of Data Streams	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5
3	45	Streaming data and cloud computing	Streaming data and cloud computing	,T- Introduction to Database Mana,T-Database Management Systems,T-Database System Concepts,T-Fundamentals of Database Syste,R-An Introduction to Database Ma,R-Principles of distributed data	Activity,Case Study,Flipped Classes,Infographics,Instructor Lead WorkShop,PPT,Professor of Practice/Adjunct Faculty/Visiting Professor,Reports,Simulation,Video Lecture	CO5

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	20EP02	External Theory	60
2	20EP02	Attendance Marks	2
3	20EP02	Mid-Semester Test-1	20
4	20EP02	Short Term Paper / Research Paper	12
5	20EP02	Mid-Semester Test-2	20
6	20EP02	Quiz	6

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	NA	NA	1	2	NA	NA	NA	NA	NA	NA	NA	NA	1
CO2	3	NA	NA	1	1	NA	NA	NA	NA	NA	NA	NA	NA	2
CO3	3	1	2	2	2	NA	NA	NA	NA	NA	NA	NA	NA	2
CO4	2	3	NA	2	2	NA	NA	NA	NA	NA	NA	NA	NA	3
CO5	3	NA	3	2	3	NA	NA	NA	NA	NA	NA	NA	NA	3
Target	2.6	2	2.5	1.6	2	NA	NA	NA	NA	NA	NA	NA	NA	2.2